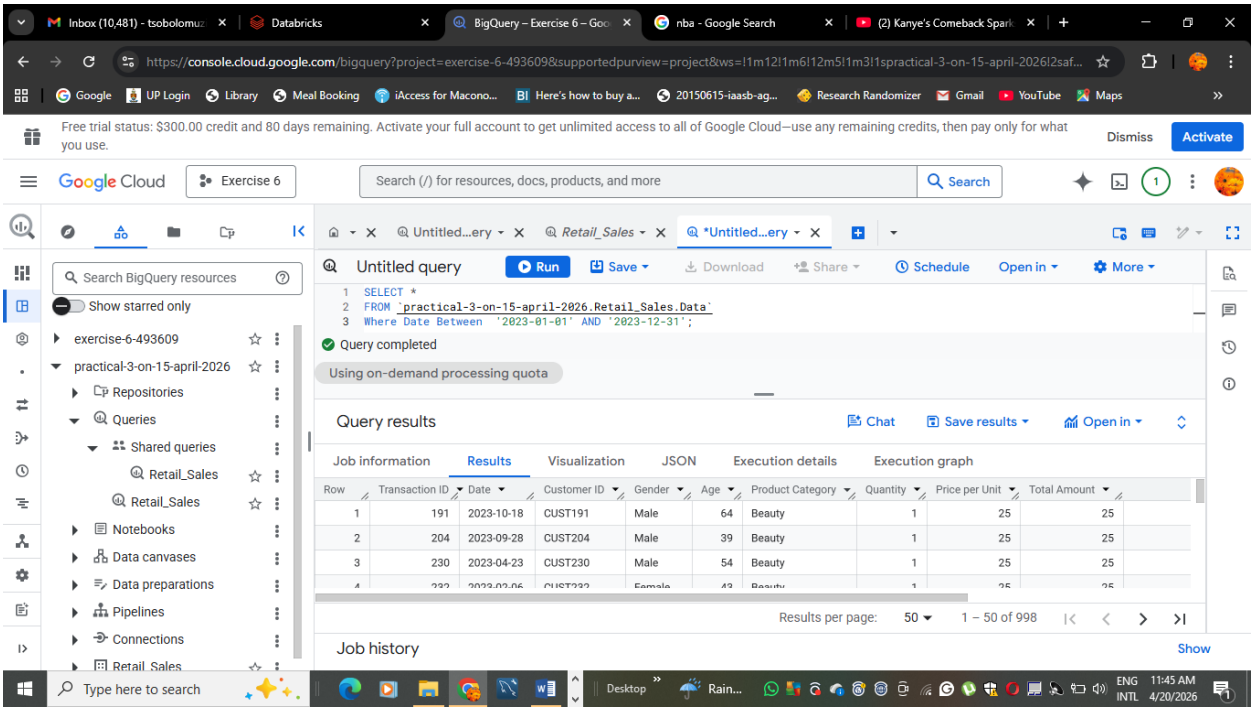
Practical Exercise

Google BigQuery

Question 1: Filter all transactions that occurred in the year 2023. Expected Output: All columns.

Query: SELECT *
FROM `practical-3-on-15-april-2026.Retail_Sales.Data`
Where Date Between '2023-01-01' AND '2023-12-31';

Answer:



Question 2: Display all transactions where the Total Amount is more than the average Total Amount of the entire dataset.
Expected Output: All columns

Query:
SELECT *
 FROM `practical-3-on-15-april-2026.Retail_Sales.Data`
 Where `Total Amount` > (SELECT AVG(`Total Amount`)
 FROM `practical-3-on-15-april-2026.Retail_Sales.Data`);

Answer:

Untitled query

```
SELECT *
FROM `practical-3-on-15-april-2026.Retail_Sales.Data`
Where `Total Amount` > (SELECT AVG(`Total Amount`)
FROM `practical-3-on-15-april-2026.Retail_Sales.Data`);
```

Query completed

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Query results

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
1	21	2023-01-14	CUST021	Female	50	Beauty	1	500	500
2	28	2023-04-23	CUST028	Female	43	Beauty	1	500	500
3	128	2023-07-05	CUST128	Male	25	Beauty	1	500	500
4	220	2023-03-03	CUST220	Male	64	Beauty	1	500	500

<u>Question 3: Calculate the total revenue (sum of Total Amount). Expected Output: Total_Revenue</u>

Query: SELECT SUM(`Total Amount`) AS Total_Revenue
FROM `practical-3-on-15-april-2026.Retail_Sales.Data`;

Answer:

Untitled query

```
SELECT SUM(`Total Amount`) AS Total_Revenue
FROM `practical-3-on-15-april-2026.Retail_Sales.Data`;
```

Query completed

Using on-demand processing quota

Query results

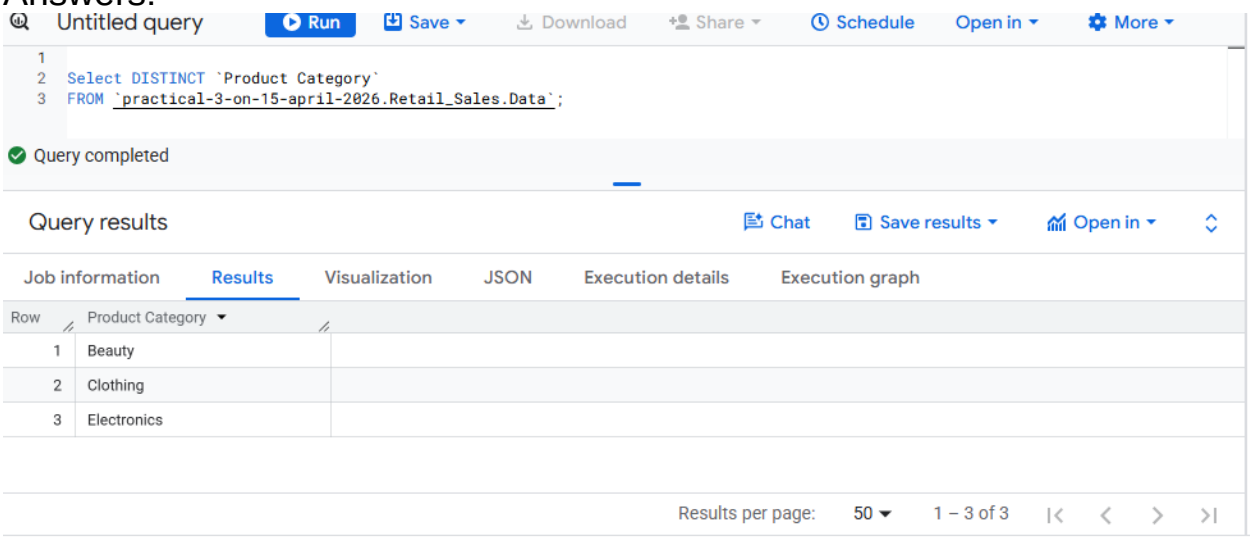
Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Total_Revenue
1	456000

Question 4: Display all distinct Product Categories in the dataset. Expected Output: Product_Category

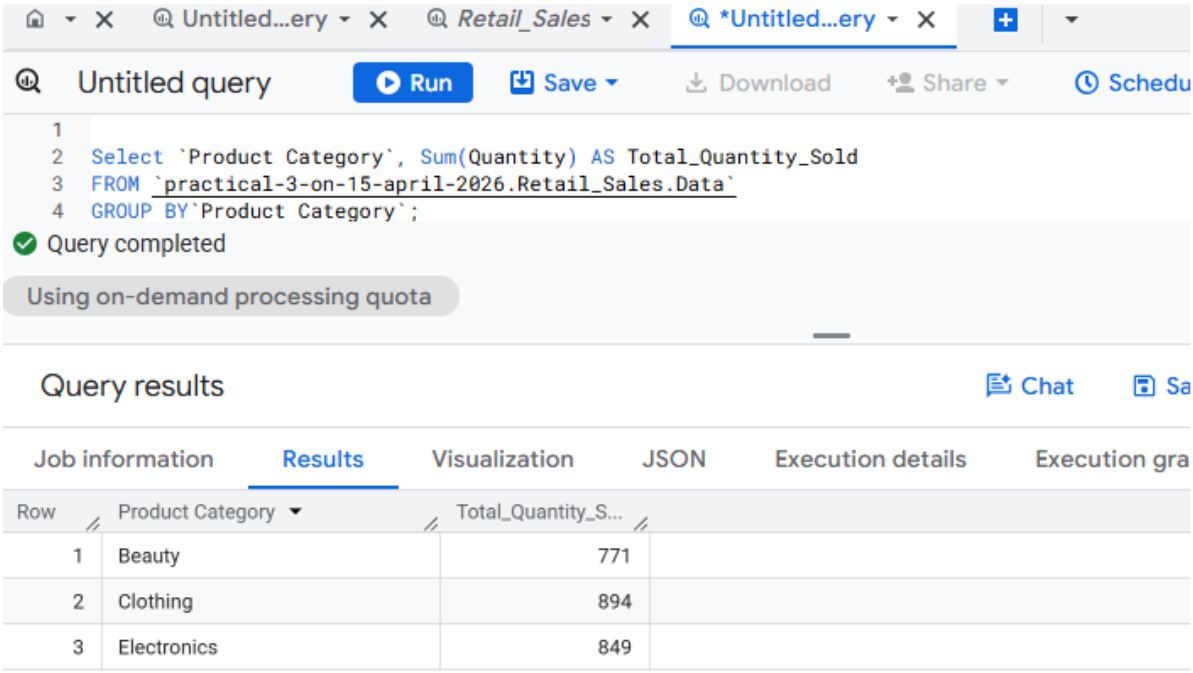
Query: Select DISTINCT `Product Category`
FROM `practical-3-on-15-april-2026.Retail_Sales.Data`;

Answers:



Question 5: For each Product Category, calculate the total quantity sold.
Expected Output: Product_Category, Total_Quantity
Query:
Select `Product Category`, Sum(Quantity) AS Total_Quantity_Sold
FROM `practical-3-on-15-april-2026.Retail_Sales.Data`
GROUP BY`Product Category`;

Answer:



Question 6: Create a column called Age_Group that classifies customers as: – 'Youth' (<30) – 'Adult' (30–59) – 'Senior' (60+)
Expected Output: Customer_ID, Age, Age_Group

Query:

```
Select `Customer ID`, Age,
  Case
      When Age < 30 Then 'Youth'
      When Age Between 30 AND 59 Then 'Adult'
      When Age > 60 Then 'Senior'
  End AS Age_Group
From `practical-3-on-15-april-2026.Retail_Sales.Data`;
```

Answer:

Untitled query

```
Select `Customer ID`, Age,
  Case
      When Age < 30 Then 'Youth'
      When Age Between 30 AND 59 Then 'Adult'
      When Age > 60 Then 'Senior'
  End AS Age_Group
From `practical-3-on-15-april-2026.Retail_Sales.Data`;
```

This query will process 16.6 KB when run.

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Query results

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Product Category	Total_Quantity_S...
1	Beauty	771
2	Clothing	894

Question 7: For each Gender, count how many high-value transactions occurred (where Total Amount > 500).
Expected Output: Gender, High_Value_Transactions

Query: Select Gender, Count(`Transaction ID`) AS High_Value_Transactions
FROM `practical-3-on-15-april-2026.Retail_Sales.Data`
WHERE `Total Amount` >500
GROUP BY Gender;

Answer:

Untitled query

```
Select Gender, Count(`Transaction ID`) AS High_Value_Transactions
FROM `practical-3-on-15-april-2026.Retail_Sales.Data`
WHERE `Total Amount` >500
GROUP BY Gender;
```

Query completed

Using on-demand processing quota

Query results

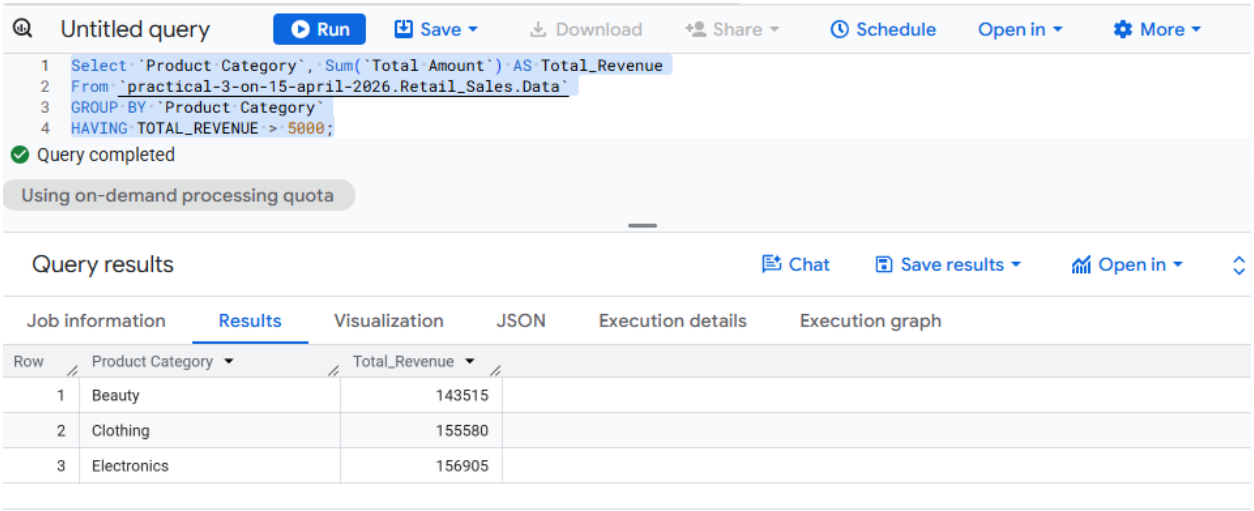
Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Gender	High_Value_Tran...
1	Female	155
2	Male	144

Question 8: For each Product Category, show only those categories where the total revenue exceeds 5,000.
Expected Output: Product_Category, Total_Revenue

Query:
```
Select `Product Category`, Sum(`Total Amount`) AS Total_Revenue
From `practical-3-on-15-april-2026.Retail_Sales.Data`
GROUP BY `Product Category`
HAVING TOTAL_REVENUE > 5000;
```

Answer:



Row	Product Category	Total_Revenue
1	Beauty	143515
2	Clothing	155580
3	Electronics	156905

Question 9: Display a new column called Unit_Cost_Category that labels a transaction as: – 'Cheap' if Price per Unit < 50
– 'Moderate' if Price per Unit between 50 and 200
– 'Expensive' if Price per Unit > 200
Expected Output: Transaction_ID, Price_per_Unit, Unit_Cost_Category

Query:
```
Select `Transaction ID`, `Price Per Unit`,
  CASE
    When `Price per Unit` < 50 then 'Cheap'
    When `Price per Unit` Between 50 AND 200 then 'Moderate'
    When `Price per Unit` > 200 then 'Expensive'
 END AS Unit_Cost_Categpry
From `practical-3-on-15-april-2026.Retail_Sales.Data`;
```

Answer:

Untitled query

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```
Select `Transaction ID`, `Price Per Unit`,
  CASE
    When `Price per Unit` < 50 then 'Cheap'
    When `Price per Unit` Between 50 AND 200 then 'Moderate'
    When `Price per Unit` > 200 then 'Expensive'
  END AS Unit_Cost_Categpry
From `practical-3-on-15-april-2026.Retail_Sales.Data`;
```

Query completed

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Query results

Chat Save results Open in

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Transaction ID	Price Per Unit	Unit_Cost_Categpry
1	191	25	Cheap
2	204	25	Cheap
3	230	25	Cheap

Question 10: Display all transactions from customers aged 40 or older and add a column Spending_Level showing:
– 'High' if Total Amount > 1000
– 'Low' otherwise
Expected Output: Customer_ID, Age, Total_Amount, Spending_Level

Query:

```
SELECT `Customer ID`, Age, `Total Amount`,
    CASE
      WHEN `Total Amount` > 1000 THEN 'High'
     ELSE 'Low'
    END AS Spending_Level
 From `practical-3-on-15-april-2026.Retail_Sales.Data`
 Where Age >= 40;
```

Answer:

Untitled query

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```sql
SELECT `Customer ID`, Age, `Total Amount`,
    CASE
      WHEN `Total Amount` > 1000 THEN 'High'
      ELSE 'Low'
    END AS Spending_Level
From `practical-3-on-15-april-2026.Retail_Sales.Data`
Where Age >= 40;
```

Query completed

Using on-demand processing quota

Query results

Chat | Save results

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Customer ID	Age	Total Amount	Spending_Level
1	CUST191	64	25	Low
2	CUST230	54	25	Low